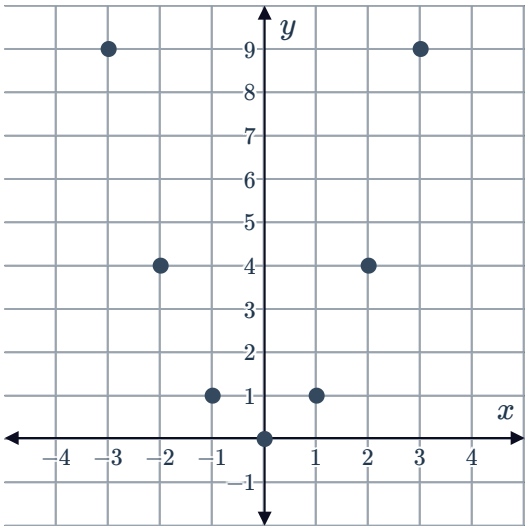
Graphing quadratic functions

1

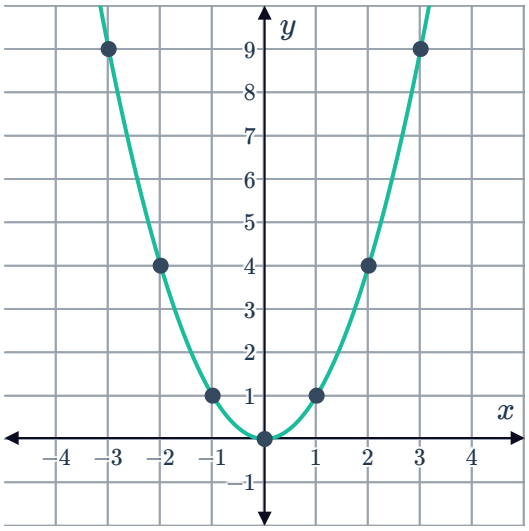
a

x	-3	-2	-1	0	1	2	3
y	9	4	1	0	1	4	9

b



c



d No

e $x = 0$

f 0

g 2

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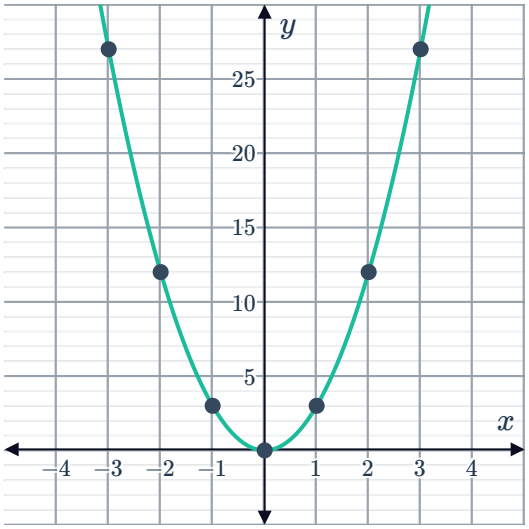
3

a

i

x	-3	-2	-1	0	1	2	3
y	27	12	3	0	3	12	27

ii

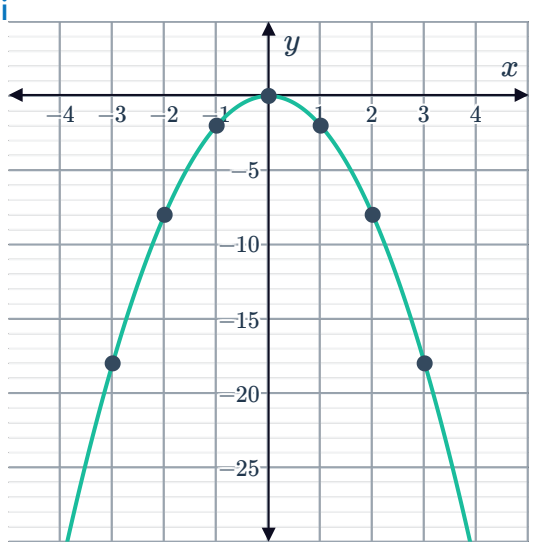


b

i

x	-3	-2	-1	0	1	2	3
y	-18	-8	-2	0	-2	-8	-18

ii



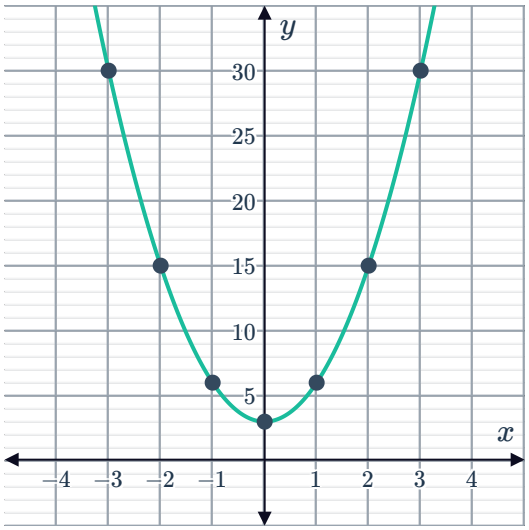
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c

i

x	-3	-2	-1	0	1	2	3
y	30	15	6	3	6	15	30

ii

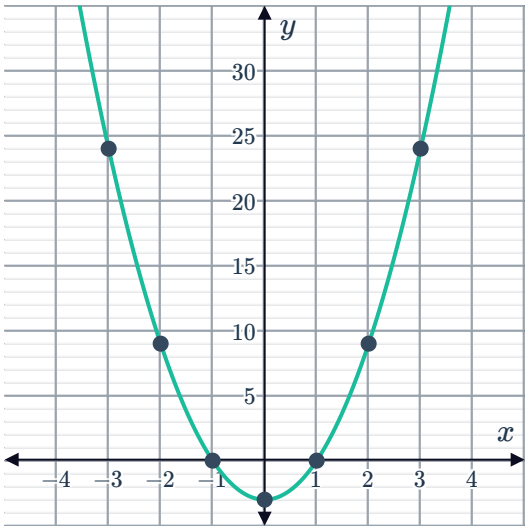


d

i

x	-3	-2	-1	0	1	2	3
y	24	9	0	-3	0	9	24

ii

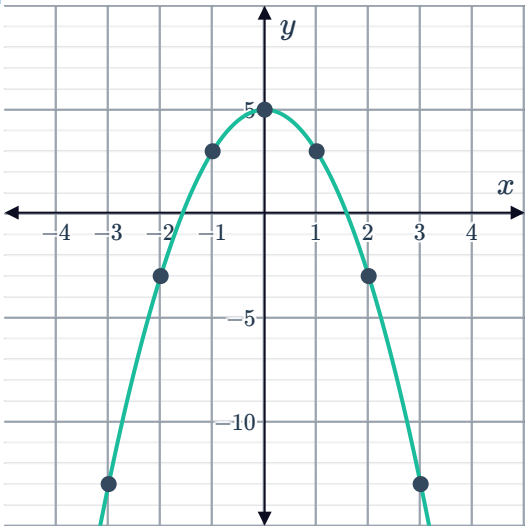


e

i

x	-3	-2	-1	0	1	2	3
y	-13	-3	3	5	3	-3	-13

ii



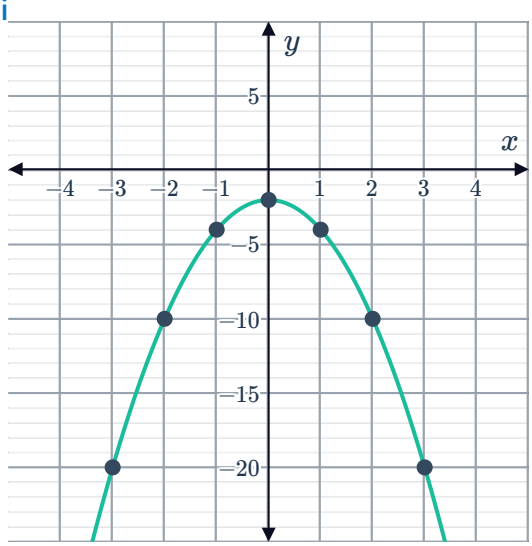
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f

i

x	-3	-2	-1	0	1	2	3
y	-20	-10	-4	-2	-4	-10	-20

ii



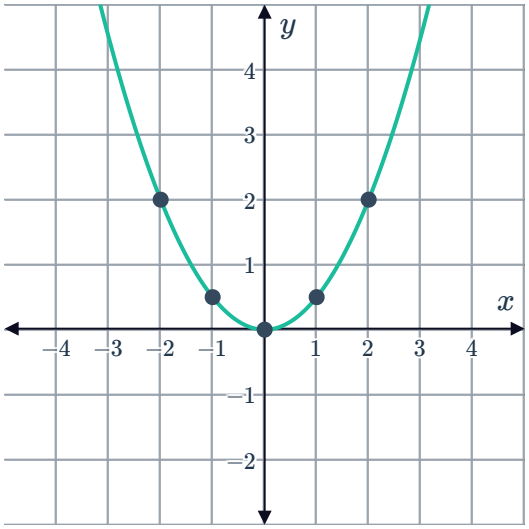
4

a

i

x	-2	-1	0	1	2
y	2	$\frac{1}{2}$	0	$\frac{1}{2}$	2

ii



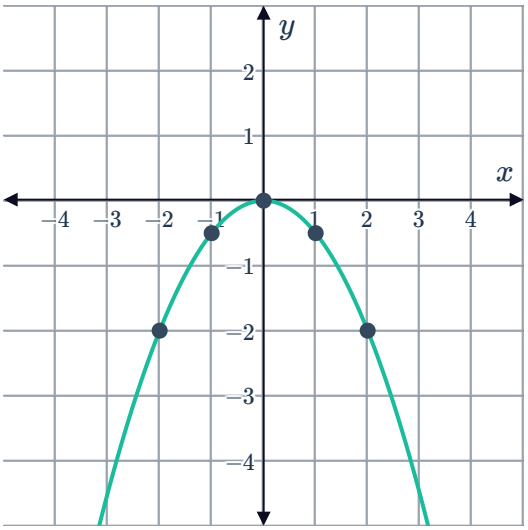
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b

i

x	-2	-1	0	1	2
y	-2	$-\frac{1}{2}$	0	$-\frac{1}{2}$	-2

ii

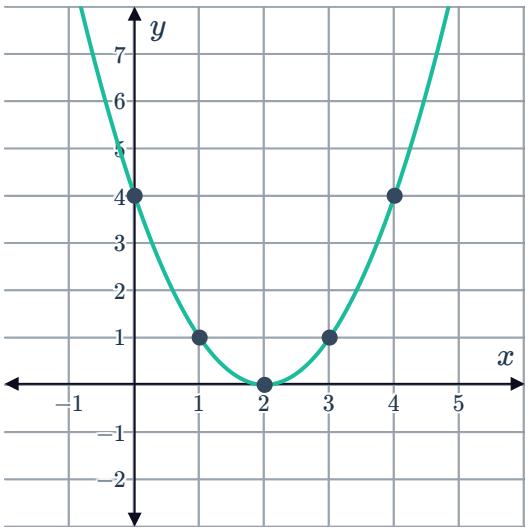


5

a

x	0	1	2	3	4
y	4	1	0	1	4

b



c 0

d 2

e $(2, 0)$

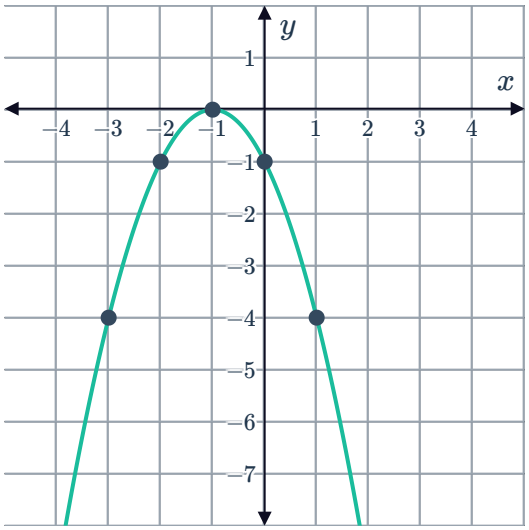
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b

i

x	-3	-2	-1	0	1
y	-4	-1	0	-1	-4

ii



iii 0

iv -1

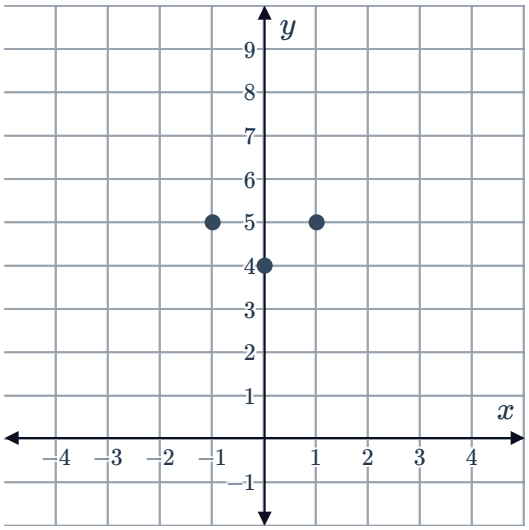
v $(-1, 0)$

8 True

9

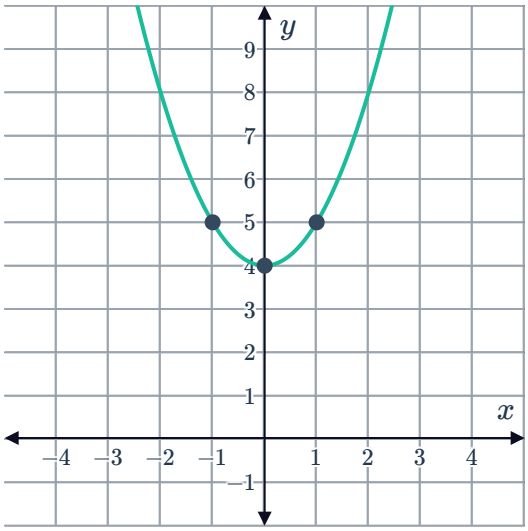
a $A(-3, 13)$, $B(-2, 8)$, $C(-1, 5)$, $D(0, 4)$, $E(1, 5)$, $F(2, 8)$, $G(3, 13)$

b



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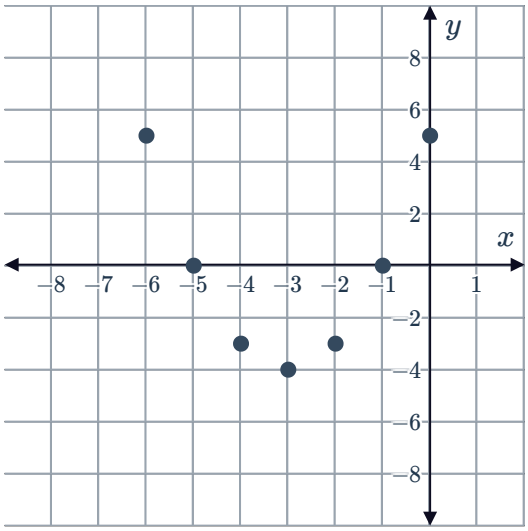
c



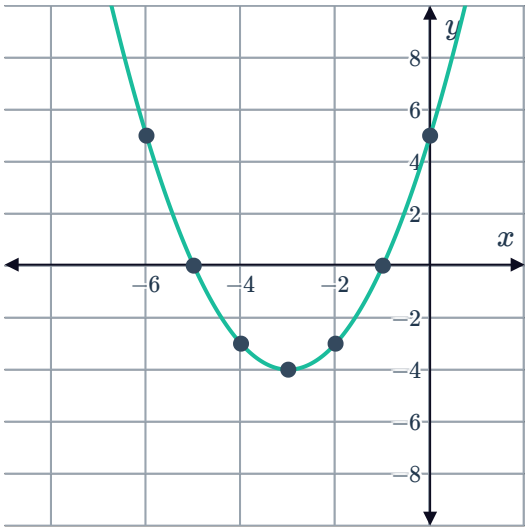
10

a $A(-5,0)$, $B(-2,-3)$, $C(-6,5)$, $D(-1,0)$, $E(0,5)$, $F(-3,-4)$, $G(-4,-3)$

b



c



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11 $y = 0$

12 Maximum value

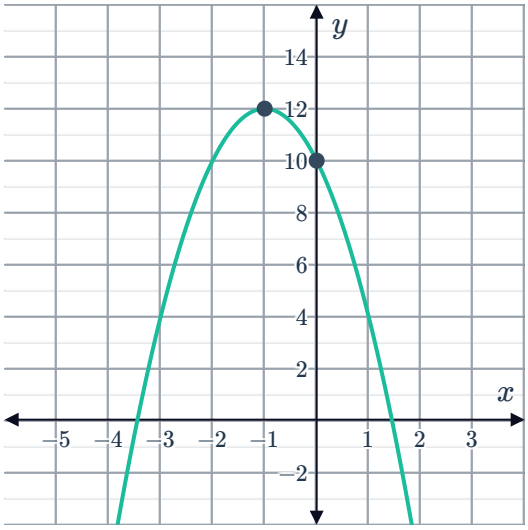
13

a $x = -1$

b 12

c $(-1, 12)$

d

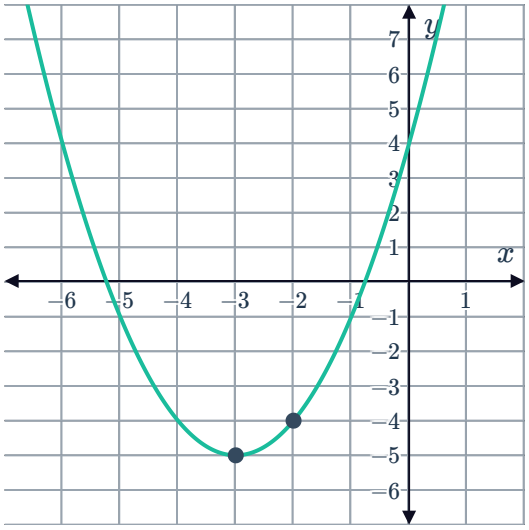


14

a $x = -3$

b -5

c



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15

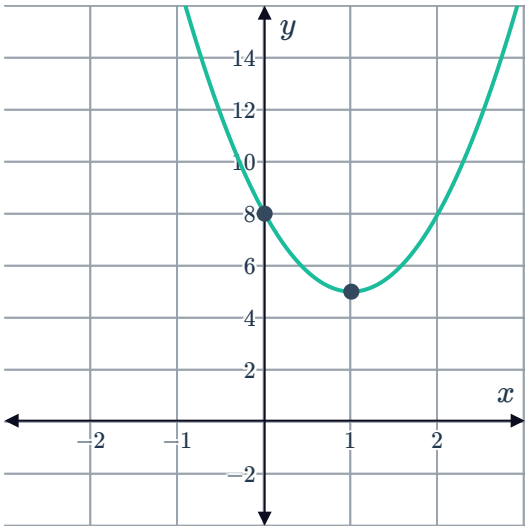
a

i $x = 1$

ii $(1, 5)$

iii 8

iv



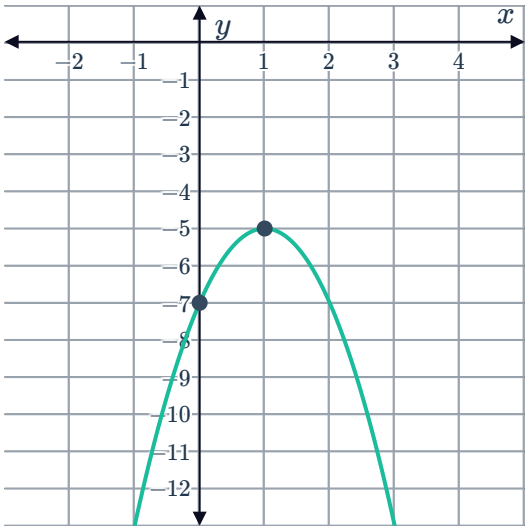
b

i $x = 1$

ii $(1, -5)$

iii -7

iv



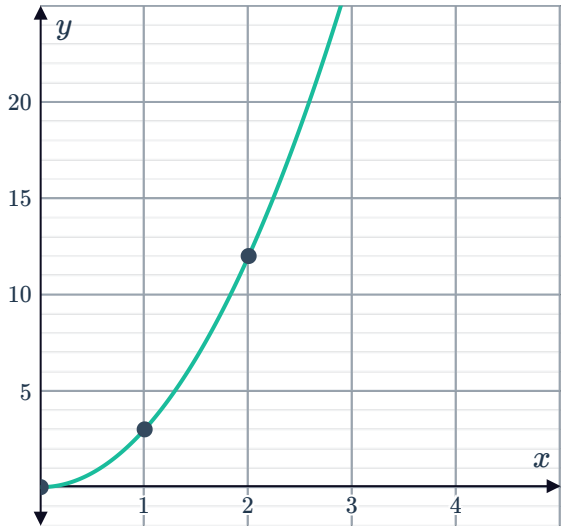
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16

a

x (m)	0	1	2	3	4
y (m)	0	3	12	27	48

b



c 0 m

d 0

e $(0, 0)$

17 $g(x)$ has the smallest minimum value which is -16

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